

CIP scenario — meeting brief for Kato-sensei

2026-07-08 Aiko branch hymeko-neuro-migration

One line. *DirectLiNGAM / CIP running over the machine-verified HyMeKo hypergraph substrate — demonstrated end-to-end on coin-toss (de-risk) and on your MetaWorld coffee-push scenario, with every discovered causal graph cross-view-verified against the HyMeKo engine.*

1. The idea we are testing (the joint)

The runtime monitor judges the task (reward-independent); CIP prioritizes reward $\leftrightarrow$ monitor disagreement; DirectLiNGAM proposes a causal graph over the rollout variables; **that graph is declared as a .hymeko signed hypergraph and the HyMeKo engine re-derives its tensor view + a Blake3 canonical hash** — so *the causal model the agent uses is provably the one a human audits* (Kato–LiNGAM joint #1). The science extension on top is **LiNGAM-SH** (signed-hypergraph LiNGAM) — the contribution proper, not integration glue.

rollouts $\rightarrow$ HyMeKo monitor $\rightarrow$ CIP variables $\rightarrow$ DirectLiNGAM $\rightarrow$ causal DAG
DAG $\rightarrow$ declare as .hymeko $\rightarrow$ engine star/clique tensor + Blake3 hash $\rightarrow$ cross-view:
declared $\equiv$ tensor

2. What is built and verified (this week)

Layer	Status	Evidence
Runtime monitors (reward-independent)	built	<code>task_monitor/</code> (coin, coffee-push, dial-turn)
CIP-export bridge (verdict $\rightarrow$ CIP vars)	built	<code>cip_export.py</code>
DirectLiNGAM consumer (numpy/scipy, no lingam)	built	recovers synthetic ground truth, recall 1.00
.hymeko declaration + engine cross-view	built	passed in 100% of runs
Coin-toss PoC (de-risk)	done	<code>contact_score</code> $\rightarrow$ <code>total_reward</code> ; farming candidate
Contact-reward ablation (intervention)	done	removing contact terms collapses the edge
MetaWorld coffee-push / dial-turn (template)	done	recovers the scenario story
MetaWorld coffee-push (real env) + multi-seed	done	reward is proximity-shaped (5/5 batches)

3. Results across the scenario

Coin-toss (de-risk). `contact_score` $\rightarrow$ `total_reward` (+0.69) is the strongest edge and `total_reward` is disconnected from `delivery_score` $\Rightarrow$ **reward_farming_candidate**. The *intervention* (Stage A, no training): recomputing the reward with the contact terms removed **collapses that edge to 0.0** and re-parents the reward onto `delivery_score` (+0.88). Supported at the reward-computation level. We did **not** “solve” coin-toss (a physics wall and an imitation gap remain); CIP *diagnosed* it.

MetaWorld coffee-push (your scenario), multi-seed 5×80 , no training:

- `near_fraction` $\leftrightarrow$ `total_reward`: **CONFIRMED stable** — present 5/5, $|w|$ median **0.973**, IQR [0.962, 0.981]. The real MetaWorld reward is **proximity/contact-shaped**. (Honest: LiNGAM flips the direction 3/2 across batches — coupling solid, direction not uniquely resolved.)
- `progress_score` $\rightarrow$ `total_reward`: **rejected** as a direct edge (present 1/5) — `near_fraction` mediates it. *The single run suggested this edge; multi-seed removed it — the value of aggregation.*
- Cross-view verification **passed in all 5/5 batches**; pass-rate median 0.51, disagreement 0.35.

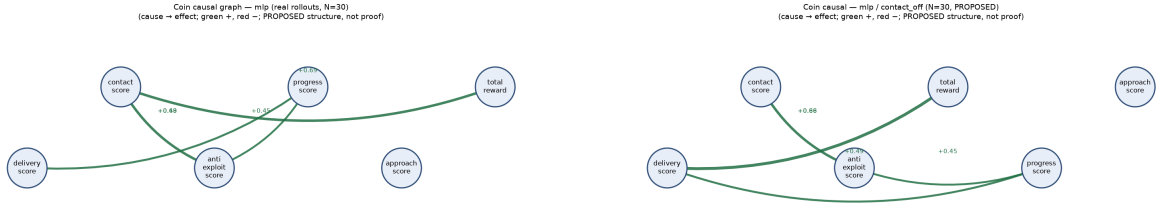


Figure 1: Coin-toss: (left) `contact_score` $\rightarrow$ `total_reward` discovered; (right) after removing the contact reward terms the edge collapses and the reward re-parents onto `delivery_score` (+0.88).

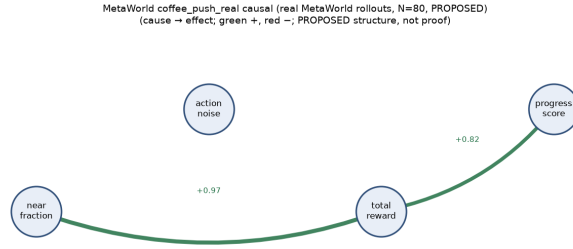


Figure 2: Real MetaWorld coffee-push: `near_fraction` $\rightarrow$ `total_reward` (+0.97, stable across 5 seeds), `progress_score` feeding reward via proximity; `action_noise` isolated (scripted policy is noise-robust). PROPOSED; cross-view-verified.

4. Behaviour (GIF stills; full GIFs in reports/gifs/metaworld_coffee_push/)

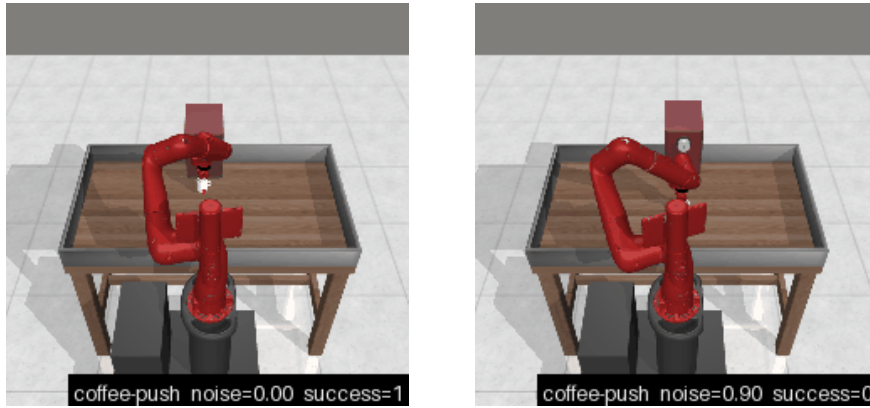


Figure 3: Scripted `SawyerCoffeePushV3Policy` in the real MetaWorld env: (left) clean, noise 0.0 $\rightarrow$ success; (right) noisy, noise 0.9 $\rightarrow$ miss. Action noise is the observed input the CIP run varies.

5. Honest scope (say this in the meeting)

- Every causal graph is **PROPOSED**; controlled ablation decides. The cross-view proves *representation* consistency (declared $\equiv$ tensor $\equiv$ hash), **not** causal truth.
- MetaWorld’s env randomization is **not controlled by the seed** — single-run order is a point estimate; the coffee-push claim rests on the 5-batch aggregate, not one run.
- Real-env is **coffee-push only** so far (clean obs $\rightarrow$ monitor mapping); dial-turn real-env needs a dial-angle extraction (its template is done).
- **No RL training** was run — all read-only rollouts of cached / scripted policies.

6. Proposed next steps (decisions for Kato)

1. **LiNGAM-SH** — constrain LiNGAM’s B to factor through the signed incidence ($B = A_{\text{out}} \Sigma A_{\text{in}}^{\top}$) so non-Gaussianity identifies the *grouping* (the theorem). Nearest prior art (arXiv:2511.03831, CAM→acyclic hypergraphs, Gaussian) validates the direction, does not pre-empt (no LiNGAM/ICA regime, no signs).
2. **Real-env reward ablation on MetaWorld** — transfer the coin Stage-A method: decompose the reward, recompute contact-off offline, check the $\rightarrow$ **total_reward** edge collapses.
3. **Multi-seed everywhere + dial-turn real-env** for a publishable ranking claim.

Environment. metaworld 3.0.0, mujoco 3.10.0 (pinned), gymnasium 1.3.0, scipy 1.17.1; native **hymeko** engine built via maturin. 90 CIP tests pass; ruff/radon/mypy(strict) clean. Commits **a39acbb**, **b3d56d3**, **d9a436f**, **52e3af9**. Backing reports: 2026-07-08-cip-phase2-coin-poc.md, 2026-07-08-cip-contact-reward-ablation-setup.md, 2026-07-08-cip-metaworld-2026-07-08-cip-metaworld-coffee-push-multiseed.md.